AiEO Visibility Challenge
Technation Canada

Team: AI Accuracy
Benonia, Chandhu, Maria, Srivani
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Challenge:

Generative AI is increasingly the primary gateway for Canadians to access information about organizations, programs, and communities. Visibility within AI systems shapes what is seen, trusted, and accessed.

Key Issues:

AI responses often show gaps in coverage of Canadian organizations.
Misinformation or incomplete information can skew public understanding.
Unequal visibility impacts access to jobs, funding, training, and public services.

Outcome of Challenge (Stage 1 and Stage 2):

Better understand AI visibility patterns and propose interventions to improve representation, accuracy, and fairness.

Research Approach & Methodology:
Designed 100 questions across the following categories

- Technology & Infrastructure (Open-source data, rural connectivity, AI/ML programs, and innovation research).
- Workforce, Skills & Education (Coding bootcamps, job training, scholarships, and academic mentorship).
- Business & Entrepreneurship (Startup funding, incubators, accelerators, and small business support).
- Indigenous Community & Economic Support (Indigenous-led organizations, housing, infrastructure, and entrepreneurship).
- Settlement & Newcomer (Immigrant services, language training, and refugee community support).
- Health, Wellness & Social Services (Mental health, healthcare, emergency aid, and housing).
- Equity & Inclusive Advocacy (Women in STEM, youth, Francophones, and disabled communities).

Data Collection:

- Ran all 100 questions across 4 AI LLM models namely, Gemini, ChatGPT, Perplexity and Claude and collected and stored raw responses in structured data sheet.

- Special care was taken to ensure clean data:
    - Unpaid vs paid subscription
    - Logged in vs not logged in
    - Browser normal vs incognito mode
    - Responses were independent of user interaction pattern learning of AI.
    - Consistent prompt for all questions across models.

Details of Data collected	Count
Number of Provinces and Territories	13
Number of Models	4
Number of Sub Categories	20
Number of questions per category	5
Retrieved top 5 organizations	5
Number of responses per models	6500
Total number of responses received	**26000**

Copy and paste the link to view dataset:

https://docs.google.com/spreadsheets/d/10_kM__M2OkEjvy5-22Y1O1lxA1mtHg-1B2IZUDIzz8I/edit?usp=sharing

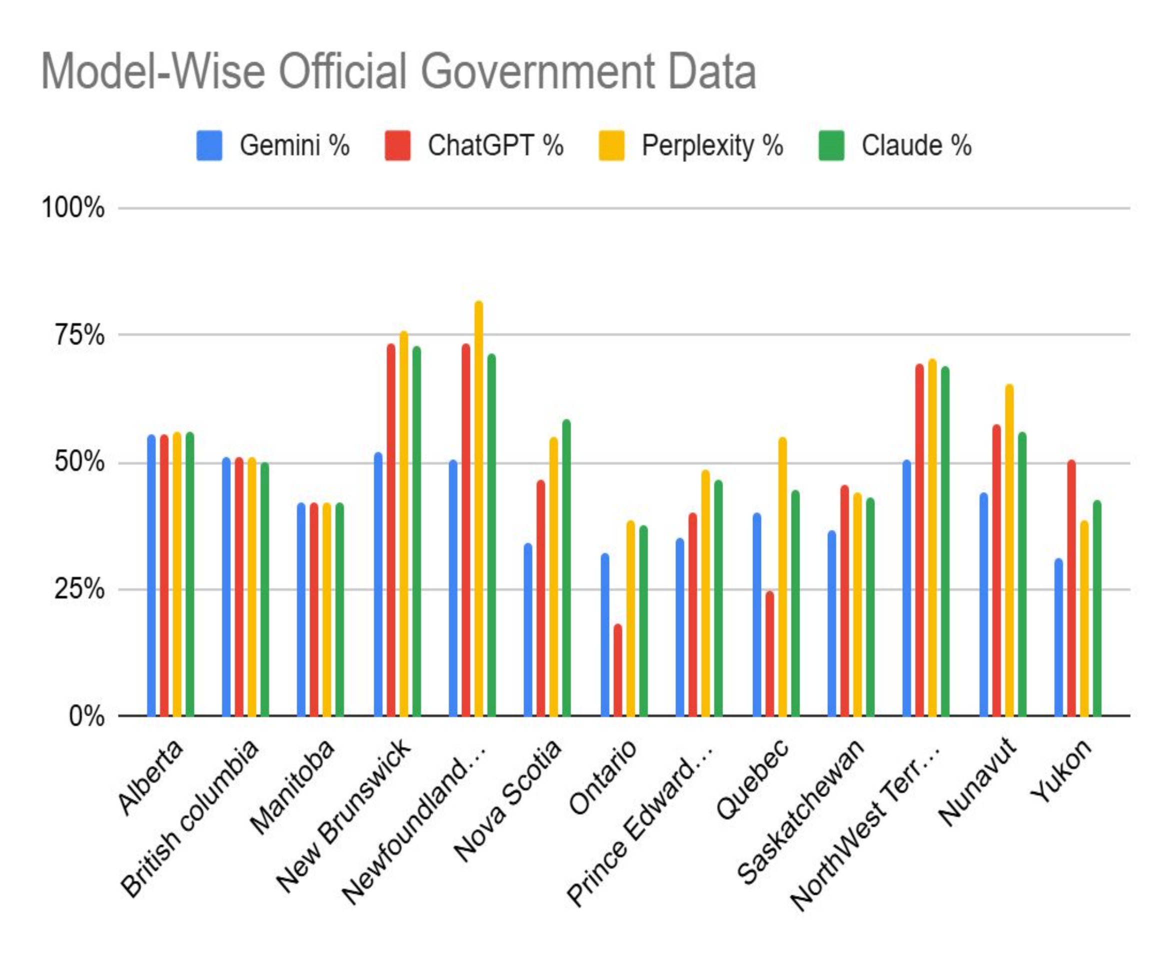
Analysis:

- To ensure the data accuracy against modals and Provincial government websites, the following set of sectors were shortlisted based on the public impact.
    - ❖ Healthcare & Mental Health Services
    - ❖ Employment & Skills Training / Workforce Development
    - ❖ Education & Student Financial Support
    - ❖ Indigenous Community Development & Services
    - ❖ Innovation, Technology & Clean Economy

- The responses returned by all the modals were crossed-verified in provincial/federal/google search to identify the % split of official and reliable govt information versus the private and NGO information provided by the modals.

Regional Performance Trends:
Models perform strongest in Newfoundland and Labrador and New Brunswick (~75–80%), while performance drops significantly in Ontario and Manitoba (below 50%), with all models also showing weaker results in Yukon.

Model Comparison:
Perplexity and Claude generally achieve the highest performance, particularly in the Atlantic provinces and territories, while Gemini shows more consistent but comparatively lower performance.



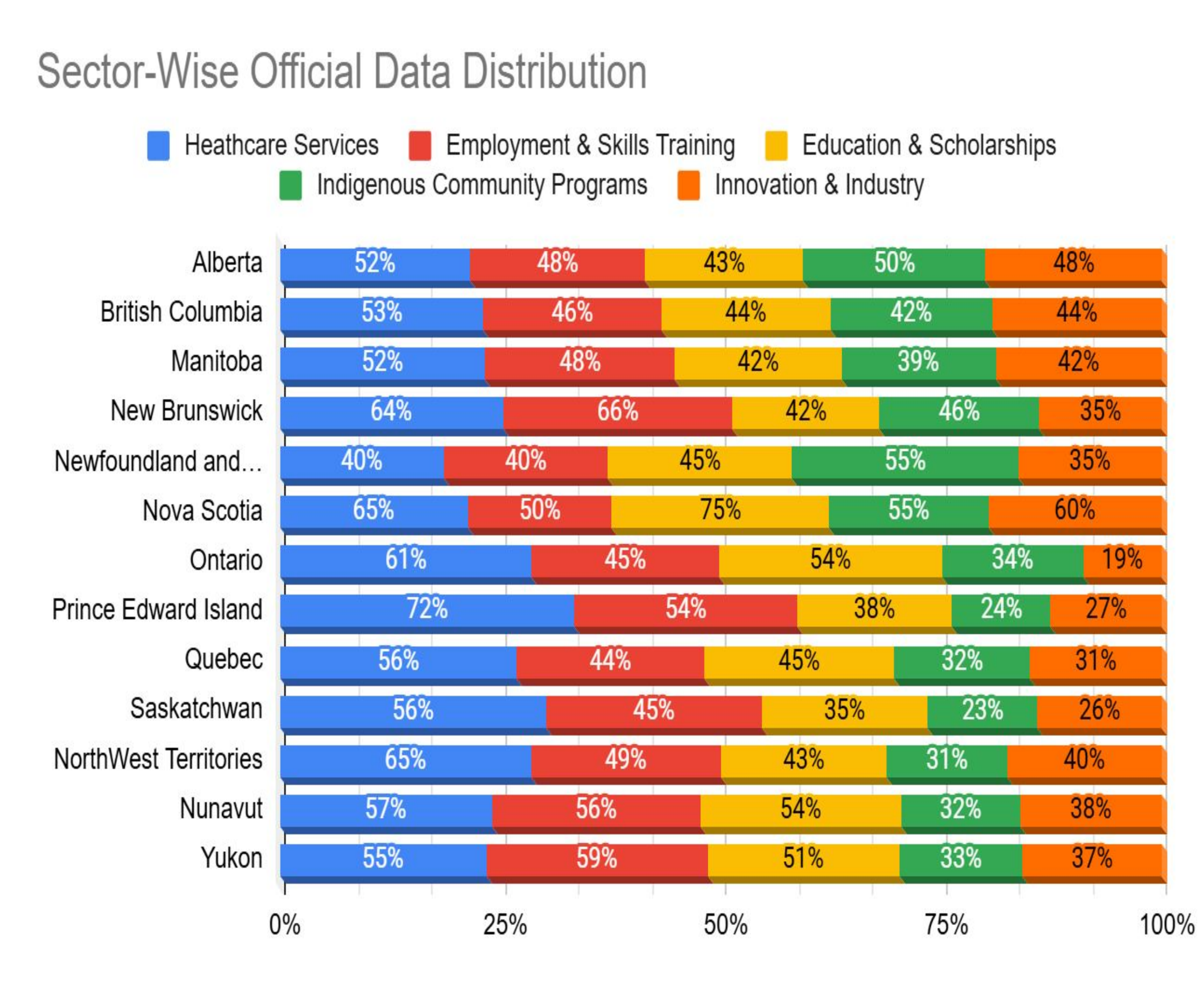
Key Observations:

- **AI Coverage:** Gemini did not give answer for ~20% of the sub-questions prompt. ChatGPT, Claude and Perplexity gave answers for 100% of the sub-questions prompts.
- **Misinformation or incomplete info:** Minor errors in service names, program eligibility, or scope. Across models, there was minimal outdated and incorrect information (~ 2 - 5%).
- **Inconsistent representation:** Some provinces have more AI coverage, compared to others.
- **Government vs. Private:** AI shows Govt data (Provincial and Federal) of approx 50 - 60% and the remaining % split across NGOs, Non-Profit and Private organizations.
- **Platform Differences:** ChatGPT vs. Claude vs. Gemini vs. Perplexity show varied accuracy and completeness, majority shows similar organizations.
- **Cross-platform differences:** Some AI models include more comprehensive community info, others rely heavily on top-ranking official sources

Sector Dominance: Healthcare Services generally occupies the largest share of data distribution in most regions, peaking at 72% in Prince Edward Island.

Regional Trends: Nova Scotia shows a uniquely high distribution for Education & Scholarships (75%), while Ontario has a notably low share for Innovation & Industry (19%) compared to its other sectors.

Indigenous Representation: Data for Indigenous Community Programs is most prominent in Newfoundland and Labrador and Nova Scotia (at 55%), but significantly lower in Saskatchewan (23%) and PEI (24%).



Implications:

- Unequal visibility can affect public access to services, training, and funding opportunities.
- Digital inequity risks widening if not addressed.

Early Insights:

- ❖ Most visible: Government health, education, and major employment programs.
- ❖ Least visible: Small regional NGOs, emerging tech initiatives.
- ❖ Misinformation patterns: Often due to outdated and the limited availability of AI training data.
- ❖ Improving AI representation supports fairness, access, and equitable knowledge distribution across Canadian organizations, sectors, communities, and programs.

Conclusion of Stage 1:

- Provinces such as Nova Scotia, New Brunswick, and Newfoundland and Labrador show the highest AI accuracy (~75%), indicating that diverse and well-balanced sector data, particularly in Education and Indigenous sectors, improves model performance.
- High data concentration in a single sector (e.g., PEI's 72% Healthcare share) does not necessarily improve AI accuracy, while limited data in key sectors (e.g., Ontario's 19% Innovation share) correlates with lower model performance, suggesting data structure and sector balance matter more than sheer volume.

Stage 2 focus: *Based on the findings from Stage 1, an appropriate solution of policy, framework & tool is planned and may be proposed in consistent with AiEO challenge's Stage 2 criteria. The planned solution has the potential to overcome the inaccuracies in the AI responses across Canadian Govt. sectors and programs.*

Thank You

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